

Your first session in R

Chapter 1 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

EXERCISE 1.1

The Nile flows are measured in units of 10^8 cubic meters. Convert the largest flow, 1,370 units, to cubic meters, and then to cubic kilometers (one cubic kilometer is 10^9 cubic meters). Do the same for the smallest flow, 456 units. What does the `e+11` in one of R's answers mean?

My code and notes

Answer

 EXERCISE 1.2

Write down the answer you expect from each line, then run the lines. Which result surprised you, and what does it tell you about the order in which R applies a minus sign and $^$?

```
2 + 3 * 4 ^ 2
(2 + 3) * 4 ^ 2
-2 ^ 2
(-2) ^ 2
```

My code and notes

Answer _____ EXERCISE 1.3

Use `Nile` and `sum()` to count the years with a flow below 700 units, below 700 or above 1,200 units, and exactly 1,000 units. Then use `!` to count the years not below 700 units, and check that this count and the first add up to 100.

My code and notes

Answer _____

 EXERCISE 1.4

Write down the value you expect `area` to have at the end, then run the lines. Explain the answer, and say how a spreadsheet would behave if `area` were a cell with a formula multiplying `width` by `height`.

```
width <- 4
height <- 3
area <- width * height
width <- 10
area
```

My code and notes

Answer _____

 EXERCISE 1.5

Use the help page for `round()` to round 1,370 to the nearest hundred, and to round 919.35 to two significant digits with `signif()`, which the same page describes. Then run `round(0.5)`, `round(1.5)`, and `round(2.5)`. What rule is R following, and in which section of the page is it explained?

My code and notes

Answer _____

 EXERCISE 1.6

Each line below stops with an error or gives an answer you should not trust. Run them one at a time, note what R says, and correct each line. The last line was meant to count the years in which the flow was above 1,000 units.

```
round(919.35 1)
mean(Nile, na.rm = true)
max(nile)
sum("Nile" > 1000)
```

My code and notes

Answer

2

How R sees data

Chapter 2 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

✎ EXERCISE 2.1

The built-in vector `month.name` holds the full names of the months. Use square brackets to get (a) the last three months, (b) every month except January and December, and (c) the months whose names have more than seven letters; `nchar(month.name)` counts the letters in each name.

My code and notes

Answer _____

✎ EXERCISE 2.2

Before you run anything, write down the type you expect from each line: `c(1, "1")`, `c(1L, TRUE)`, `c(2.5, FALSE)`, `c(NA, 3L)`, `1:3 / 2`, and `c("a", NA)`. Then check each one with `typeof()`, and explain any surprise with a rule from the chapter.

My code and notes

Answer _____

✎ EXERCISE 2.3

Using `life` and `year` from this chapter, find how many of the 12 values are above 50 and what share that is. In which year did Egypt's life expectancy first pass 50? Then compute its rise between 1977 and 2007, picking both values with conditions on `year` rather than with positions.

My code and notes

Answer _____**✎ EXERCISE 2.4**

In `penguins`, how many birds have no recorded flipper length? What is the mean flipper length of all the birds that were measured, and what is the mean for Gentoo penguins only? Use `$`, a condition on `species`, and `na.rm = TRUE`.

My code and notes

Answer _____

✎ EXERCISE 2.5

These six survey answers are invented for the exercise: `c("small", "large", "medium", "large", "small", "small")`. Turn them into a factor with the default levels and count them with `summary()`. Then make a second factor whose levels run from small to large, and count again. Run `as.integer()` on both. Why do the numbers differ when the answers are the same?

My code and notes

Answer _____**✎ EXERCISE 2.6**

Run `glimpse(gapminder)` and `summary(gapminder)`; `glimpse()` quotes factor values when a level contains a comma, so trust the `<fct>` tag. Which numeric column has the widest gap between its mean and median, relative to the median, and why? What is the smallest population, and which country and year does it belong to? Use `min()` and a condition, not sorting.

My code and notes

Answer _____

3

Getting data in and out

Chapter 3 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

✎ EXERCISE 3.1

Write `country_codes` to a CSV file and to an RDS file, and read both back. Which types did `readr` guess for the CSV copy, and how do they differ from the types in the RDS copy? Read the CSV file again with a `col_types` string that gives every column the type it has in `country_codes`, and check the number of rows against the original.

My code and notes

Answer _____

✎ EXERCISE 3.2

Read `data/gapminder.csv` with `n_max = 1000`. Does R warn you that rows were left out? Write one line of code that compares the number of rows read with the number of lines in the file and returns `FALSE` for this read.

My code and notes

Answer _____

✎ EXERCISE 3.3

Read `data/afghanistan_formatted.csv` with `col_types = "ffidid"` and save the result as `afg_ffidid`. How many problems does `readr` report, and in which columns? (`problems(afg_ffidid)$col` lists the column of every problem.) Why does `pop` fail now, when `readr`'s own guess read it without trouble? Then write a `cols()` specification, with an `na` argument, that reads the file with the types of `"ffidid"` wherever possible and no problems at all.

My code and notes

Answer _____**✎ EXERCISE 3.4**

Write the first 12 rows of `gapminder` with `write_csv2()` and look at the first three lines of the file with `read_lines()`. Read the file with `read_csv()` and then with `read_csv2()`. How many columns did `read_csv()` find, what did it make of the numbers, and why?

My code and notes

Answer _____

✎ EXERCISE 3.5

Write a workbook with three sheets named `first`, `codes`, and `last`: the first 12 rows of `gapminder`, `country_codes`, and the last 12 rows of `gapminder` (use `tail()`). List the sheets, then read the third sheet once by name and once by position. Which country does it hold? Finally, read only the cells `A1:D4` of that sheet.

My code and notes

Answer _____**✎ EXERCISE 3.6**

Write `spellings` with `write_csv()`, which uses UTF-8, and read the file back with `locale(encoding = "latin1")`. What happens to each accented letter, and why does each one turn into two characters? What does `guess_encoding()` say about this file?

My code and notes

Answer _____

4

Asking questions of a table

Chapter 4 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

✎ EXERCISE 4.1

Keep the `gapminder` rows for 1987 in which life expectancy was above 72 years. How many countries are there, and which continent has the most of them? Answer the second question with one more verb.

My code and notes

Answer _____

✎ EXERCISE 4.2

For a country of your choice, select `year`, `lifeExp`, and `gdpPercap`, and sort the rows from the highest GDP per capita to the lowest. In which year was GDP per capita highest? Was it also the year of the longest life expectancy?

My code and notes

Answer _____

✎ EXERCISE 4.3

Add a column `gdp_billions` that holds each country's total GDP in billions of dollars. Which five countries had the largest economies in 1952, and which five in 2007?

My code and notes

Answer _____**✎ EXERCISE 4.4**

For each continent and year, compute the median GDP per capita and the number of countries. Which continent had the highest median in 1977, was it still the highest in 2007, and how many countries does that median rest on? Use `.groups = "drop"` and explain in one sentence what it changed.

My code and notes

Answer _____

✎ EXERCISE 4.5

Using `penguins`, count the birds by species and island. Then compute the mean body mass for each combination of species and sex, skipping missing values. Which species shows the largest difference between males and females?

My code and notes

Answer _____**✎ EXERCISE 4.6**

Using `ucb` from this chapter, compute for each department the overall admission rate (men and women together) and the share of its applicants who were women. Sort the result by admission rate. What does the table show about the departments that women applied to most?

My code and notes

Answer _____

5

Cleaning data

Chapter 5 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

✎ EXERCISE 5.1

Make a logical column `full_clutch` that is `TRUE` when `clutch_completion` is “Yes”. For each species, under its short name, compute the number of birds and the percentage whose nest was never observed with a full clutch. Which species has the highest percentage?

My code and notes

Answer _____

✎ EXERCISE 5.2

What does `parse_number()` return for the three codes in `study_name`, such as “PAL0708”, and why is that a poor column for analysis? Use `case_when()` to make a column `season` with the values “2007-08”, “2008-09”, and “2009-10”, and check it against `year(date_egg)` with `count()`.

My code and notes

Answer _____

 EXERCISE 5.3

How many birds have a body mass but no `delta_15n` value? Count their comments. Which comment gives a reason that has nothing to do with blood?

My code and notes

Answer _____ EXERCISE 5.4

These answers to the question “Which country do you live in?” are invented. Clean them so that each country appears under one spelling, with the missing answer kept as `NA`. How many answers does each country have?

```
answers <- tibble(country = c(
  "Egypt", " egypt", "EGYPT ", "Kenya", "kenya", "Peru",
  "Perú", "Viet Nam", "Vietnam", "viet nam", NA))
```

My code and notes

Answer _____

 EXERCISE 5.5

This invented table holds six shipping dates from two offices. The Nairobi office writes the day first and the Houston office writes the month first. Make a column `date` that parses each row with the right function. How many different months appear? How many would appear if every row were parsed with `dmy()`?

```
shipments <- tibble(
  office = rep(c("Nairobi", "Houston"), each = 3),
  shipped = rep(c("03/04/2024", "11/05/2024",
                 "02/12/2024"), 2)
)
```

My code and notes

Answer _____

 EXERCISE 5.6

These eight thruster firings are invented. Convert every impulse to newton-seconds, at 4.448222 newtons per pound-force, and compute the total. What total would a program get if it read every value as newton-seconds, and what share of the true total would it miss?

```
firings <- tibble(  
  impulse = c(0.52, 0.48, 2.31, 0.50, 2.25, 0.47,  
              0.51, 2.28),  
  unit = c("lbf s", "lbf s", "N s", "lbf s", "N s",  
           "lbf s", "lbf s", "N s")  
)
```

My code and notes

Answer

6

Reshaping and joining tables

Chapter 6 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

✎ EXERCISE 6.1

Using `relig_long`, compute for each religion the share of its respondents who reported an income above 150,000 dollars (the band `>150k`), leaving the band `Don't know/refused` out of the total. Which religion has the highest share, and how many respondents does that share rest on? Which religion in the top ten has the fewest respondents?

My code and notes

Answer _____

✎ EXERCISE 6.2

Using `billboard_long`, find the number of weeks each song spent in the top 10 (rank 10 or better). Which song or songs spent the most weeks there, and how many of the 317 songs never reached the top 10?

My code and notes

Answer _____

✎ EXERCISE 6.3

`pivot_longer()` accepts several names in `names_to` together with a `names_sep` argument that says where to split the column names. Use them to tidy `who2` in a single call, with columns `diagnosis`, `sex`, `age`, and `cases`, and check that the result has as many rows as `who_tidy`. Then total the cases recorded in 2010 by age group and sex, and widen the result so that men and women sit side by side. Which age group had the most cases, and in which age group were there more cases among women than among men?

My code and notes

Answer _____**✎ EXERCISE 6.4**

Join `flights` to `airlines` and compute each airline's mean arrival delay in minutes, skipping missing values, together with its number of flights and the number of those flights with a recorded delay. Sort by the mean delay. Which airline, by its full name, had the longest mean delay, and on how many flights does that mean rest? Check the row counts before and after the join.

My code and notes

Answer _____

✎ EXERCISE 6.5

Make a copy of `planes` with only `tailnum` and `year`, renamed to `year_built`, and left-join it to `flights` by tail number, with `relationship = "many-to-one"`. How many flights were made by planes built before 1990? How many flights have a missing `year_built`, and what are the two different reasons a flight can end up with one? Use a filtering join to count each.

My code and notes

Answer

✎ EXERCISE 6.6

Of the 3,322 planes in `planes`, how many made at least one flight from New York in 2013, and how many made none? Use a filtering join for each and check that the two numbers add up to 3,322. Then count the flights from LaGuardia (`origin == "LGA"`) by tail number, sorted from most to fewest, and look up the three busiest tail numbers in `planes`. What do you find, and why? What does the first row of the count tell you?

My code and notes

Answer



Describing data with numbers

Chapter 7 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

EXERCISE 7.1

In `gapminder` for 1952, compute the mean and the median of `gdpPerCap`. Find the country with the highest value, then compute both summaries again without it. Which of the two moved more, and why?

My code and notes

Answer _____

EXERCISE 7.2

For each penguin species, compute the number of birds, the number of missing values, and the mean, median, standard deviation, and IQR of `bill_length_mm`. Then use the IQR rule within each species to flag birds with unusual `flipper_length_mm`. How many birds are flagged, and which species and islands do they come from?

My code and notes

Answer _____

✎ EXERCISE 7.3

For each continent in 2007, compute the mean GDP per capita of its countries and the mean weighted by population. In which continent does weighting lower the figure the most, and in which does it raise it the most? Name the country or countries that you think explain each change, and check your guess.

My code and notes

Answer _____**✎ EXERCISE 7.4**

Compute each continent's share of the total population in 1977 and in 2007 (keep two decimal places), and the percentage change in each continent's population between those two years. Which continents' shares fell even though their populations grew? Describe Europe's change in share both in percentage points and in percent.

My code and notes

Answer _____

✎ EXERCISE 7.5

In `anscombe`, set 3 has one point far from the line through the others. Find it (the row where `y3` is largest), drop it, and compute the correlation of `x3` and `y3` again. Then do the same for set 4, dropping the row where `x4` is 19, and explain each result in a sentence.

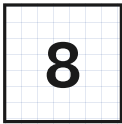
My code and notes

Answer _____**✎ EXERCISE 7.6**

Load the `datsauRus` package and look at `datasaurus_dozen`, which holds 13 datasets in three columns: `dataset`, `x`, and `y`. For each dataset, compute the number of rows, the means and standard deviations of `x` and `y`, and the correlation of `x` and `y`. What do the results tell you to do before you describe any one of these datasets in a report?

My code and notes

Answer _____



Building charts with ggplot2

Chapter 8 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

EXERCISE 8.1

Draw Figure 8.2 again for 1952 instead of 2007. One country sits far to the right of all the others. Find out with `filter()` and `arrange()` which country it is, and what its GDP per capita and life expectancy were. How does its life expectancy compare with that of the countries in the cluster at the left?

My code and notes

Answer _____

EXERCISE 8.2

Draw a line chart of life expectancy from 1952 to 2007 for the five countries of North Africa (Algeria, Egypt, Libya, Morocco, and Tunisia), with one line for each country, mapped to `color`. Which country had the lowest life expectancy in 1952, and which had the highest in 1977?

My code and notes

Answer _____

✎ EXERCISE 8.3

Count the countries in each continent in 2007 in two ways: once with `geom_bar()` on `gap_2007`, and once by computing the counts with `count()` and drawing them with `geom_col()`. Make the second chart horizontal, with the most countries at the top, and give it a title and axis titles of your own. Check your bars against the `countries` column that Chapter 4 printed.

My code and notes

Answer _____**✎ EXERCISE 8.4**

Draw box plots of life expectancy by continent for 1952. Which continents have outliers, which countries are they, and does each lie above or below its box? Check your answer with the IQR rule from Chapter 7, computed with `group_by()` and `quantile()`.

My code and notes

Answer _____

✎ EXERCISE 8.5

Draw a histogram of life expectancy in 2007 with `binwidth = 2.5`, then add `facet_wrap(~ continent, ncol = 1)`. Which continent forms the long left tail of Figure 7.1? Chapter 7 counted 19 countries with a life expectancy below 50 years; how many of them are in that continent, and which country is the exception? Check with `filter()` and `count()`.

My code and notes

Answer _____**✎ EXERCISE 8.6**

Compute the median life expectancy of each continent in 1982 and in 2007, as Chapter 4 did for 1952 and 2007. Draw those ten numbers as a dot-and-line chart like Figure 8.12, with the continents on the vertical axis and the year mapped to color. The year is stored as a number, so map `color = factor(year)`; what happens if you leave out `factor()`? Which continent has the longest line and which the shortest, and how long are they in years?

My code and notes

Answer _____



Honest charts, ready to publish

Chapter 9 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

EXERCISE 9.1

Compute the median GDP per capita of each continent in 2007 and draw it as a bar chart that starts at zero. Then draw it again with `coord_cartesian(ylim = c(5000, 31000), expand = FALSE)`. What happens to the bars for Africa and Asia, and why is that worse than an exaggerated difference? How many times as large as Asia's median is Europe's?

My code and notes

Answer _____

EXERCISE 9.2

Draw a histogram of the population of the 142 countries in 2007, first with an ordinary horizontal axis and then with `scale_x_log10()` and `label_comma()`. Which version lets you see how the populations are spread? How many countries had more than 100 million people?

My code and notes

Answer _____

✎ EXERCISE 9.3

Using `penguins`, draw a bar chart with one bar per island, filled by species, and add `position = "fill"` to `geom_bar()` so that each bar shows shares that add up to 100%. Label the vertical axis with `label_percent()`. Which island has only one species, and what share of the penguins on Dream are Chinstrap?

My code and notes

Answer _____**✎ EXERCISE 9.4**

Draw the life expectancy of every Asian country from 1952 to 2007 as gray lines, then draw two countries of your choice on top in two Okabe–Ito colors, with their names written at the end of their lines instead of a legend. Which Asian country had the lowest life expectancy in 2007?

My code and notes

Answer _____

✎ EXERCISE 9.5

Instead of a dual-axis chart, show GDP per capita and life expectancy in Egypt on one axis by dividing each by its 1952 value and multiplying by 100, so that both start at 100. Reshape the two new columns into one with `tidyr::pivot_longer()` from Chapter 6, or draw two `geom_line()` layers. By 2007, how far had each index risen, and which grew faster?

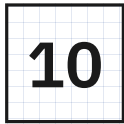
My code and notes

Answer _____**✎ EXERCISE 9.6**

Add an argument `legend = "top"` to `theme_house()` and use it for `legend.position`, so that `theme_house(legend = "none")` removes the legend. Apply it to Figure 9.8 and save the chart with `ggsave()` at 3.5 by 2.5 inches and 300 dpi. How many pixels wide and tall is the file? If you saved it as 1,050 by 750 pixels at 150 dpi instead, would the text look larger or smaller, and why?

My code and notes

Answer _____



From question to report

Chapter 10 exercises · 6 exercises · R 4.1 or later

Name _____ Date _____

✎ EXERCISE 10.1

Write the header for a report titled “Ozone in New York, summer 1973”, with your name as author and today’s date, that renders first to Word and second to a self-contained HTML file, and that hides its code. Which file does the Render button produce? Which command renders both files from the Terminal tab?

My code and notes

Answer _____

✎ EXERCISE 10.2

Add a sentence to the Findings section of the storm report that says how many storms formed in September and what percentage of them became hurricanes, with both numbers computed in a chunk and written with inline code. Render the report. What does the sentence say?

My code and notes

Answer _____

✎ EXERCISE 10.3

Split the storms into two periods, 1979–2001 and 2002–2024, with `if_else()`. For each period, compute the percentage of its storms that formed in each month, then use `pivot_wider()` from Chapter 6 to put the two periods side by side, one column each. How many storms does each period have, and did September’s share change between them?

My code and notes

Answer _____**✎ EXERCISE 10.4**

Add a chunk to the storm report that prints a table of the storms whose peak wind reached at least 160 knots, with their name, year, month, and peak wind in km/h. Give the chunk the label `tbl-strongest` and a caption, and refer to the table in the text with `@tbl-strongest`. Which storms are they, and how strong was the strongest in km/h?

My code and notes

Answer _____

✎ EXERCISE 10.5

Write a short Quarto report on `airquality`, which comes with R (see `?airquality`), that asks which month of summer 1973 had the highest median ozone reading in New York and how many days lack a reading. Include a table of the median and the missing days by month, a box plot of ozone by month, and a sentence of findings written with inline code. What are the month and the number of days?

My code and notes

Answer _____**✎ EXERCISE 10.6**

Add a check to the report from Exercise 10.5 that stops the render when any month lacks an ozone reading on more than 10 days, and render it. What happens, and which month causes it? Then replace the check with a sentence that tells the reader about that month.

My code and notes

Answer _____